	LLaMA 2 70B	GPT-3.5	Mixtral 8x7B
MMLU (MCQ in 57 subjects)	69.9%	70.0%	**70.6%**
HellaSwag (10-shot)	**87.1%**	85.5%	86.7%
ARC Challenge (25-shot)	85.1%	85.2%	**85.8%**
WinoGrande (5-shot)	**83.2%**	81.6%	81.2%
MBPP (pass@1)	49.8%	52.2%	**60.7%**
GSM-8K (5-shot)	53.6%	57.1%	**58.4%**
MT Bench (for Instruct Models)	6.86	**8.32**	8.30

Table 3: Comparison of Mixtral with Llama 2 70B and GPT-3.5. Mixtral outperforms or matches Llama 2 70B and GPT-3.5 performance on most metrics.

Evaluation Differences. On some benchmarks, there are some differences between our evaluation protocol and the one reported in the Llama 2 paper: 1) on MBPP, we use the hand-verified subset 2) on TriviaQA, we do not provide Wikipedia contexts.

3.1 Multilingual benchmarks

Compared to Mistral 7B, we significantly upsample the proportion of multilingual data during pretraining. The extra capacity allows Mixtral to perform well on multilingual benchmarks while maintaining a high accuracy in English. In particular, Mixtral significantly outperforms Llama 2 70B in French, German, Spanish, and Italian, as shown in Table 4.

	Active	French			German			Spanish			Italian		
Model	**Params**	Arc-c	HellaS	MMLU	Arc-c	HellaS	MMLU	Arc-c	HellaS	MMLU	Arc-c	HellaS	MMLU
LLaMA 1 33B	33B	39.3%	68.1%	49.9%	41.1%	63.3%	48.7%	45.7%	69.8%	52.3%	42.9%	65.4%	49.0%
LLaMA 2 70B	70B	49.9%	72.5%	64.3%	47.3%	68.7%	64.2%	50.5%	74.5%	66.0%	49.4%	70.9%	65.1%
Mixtral 8x7B	13B	**58.2%**	**77.4%**	**70.9%**	**54.3%**	**73.0%**	**71.5%**	**55.4%**	**77.6%**	**72.5%**	**52.8%**	**75.1%**	**70.9%**

Table 4: Comparison of Mixtral with Llama on Multilingual Benchmarks. On ARC Challenge, Hellaswag, and MMLU, Mixtral outperforms Llama 2 70B on 4 languages: French, German, Spanish, and Italian.

3.2 Long range performance

To assess the capabilities of Mixtral to tackle long context, we evaluate it on the passkey retrieval task introduced in [23], a synthetic task designed to measure the ability of the model to retrieve a passkey inserted randomly in a long prompt. Results in Figure 4 (Left) show that Mixtral achieves a 100% retrieval accuracy regardless of the context length or the position of passkey in the sequence. Figure 4 (Right) shows that the perplexity of Mixtral on a subset of the proof-pile dataset [2] decreases monotonically as the size of the context increases.

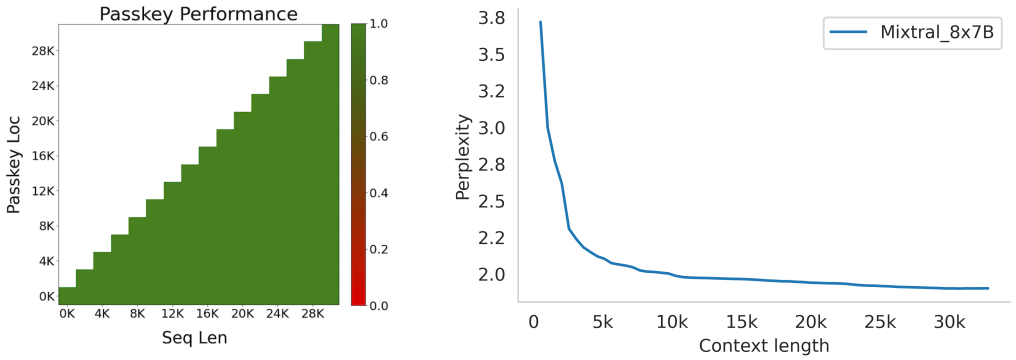


Figure 4: Long range performance of Mixtral. (Left) Mixtral has 100% retrieval accuracy of the Passkey task regardless of the location of the passkey and length of the input sequence. (Right) The perplexity of Mixtral on the proof-pile dataset decreases monotonically as the context length increases.